

Challenge (Habanero)

- Q1.** Solve the inequality $2x - 5 > 11$ for x . What is the solution set?
- (A) $x > 8$
(B) $x < 8$
(C) $x \geq 8$
(D) $x \leq 8$
(E) $x = 8$
- Q2.** A gamer's score s must satisfy $s + 20 > 100$ to reach the next level. What is the minimum score required?
- (A) $s > 80$
(B) $s < 80$
(C) $s = 80$
(D) $s \geq 80$
(E) $s \leq 80$
- Q3.** The inequality $x - 3 < 7$ represents the allowable delay x (in minutes) for a gamer's connection. What interval represents the solution on a number line?
- (A) $(-\infty, 10)$
(B) $(-\infty, 4)$
(C) $[4, \infty)$
(D) $(-4, \infty)$
(E) $(-10, 4)$
- Q4.** In a coding challenge, the player's time t (in seconds) is constrained by $t + 5 > 30$. What is the solution to this inequality?
- (A) $t > 25$
(B) $t < 25$
(C) $t = 25$
(D) $t \geq 25$
(E) $t \leq 25$
- Q5.** A gamer's rank is determined by the inequality $r - 2 > 15$, where r is the rank. What is the minimum rank?
- (A) $r > 17$
(B) $r < 17$
(C) $r = 17$
(D) $r \geq 17$
(E) $r \leq 17$
- Q6.** In a data usage scenario, the inequality $d + 10 < 50$ represents the data d (in MB) a user can upload. What interval notation represents the solution set?
- (A) $(-10, 40)$
(B) $(-\infty, 40)$
(C) $[40, \infty)$
(D) $(-40, \infty)$
(E) $(-40, 10)$
- Q7.** A game developer's profit p is given by $p - 15 > 200$. What is the minimum profit?
- (A) $p > 215$
(B) $p < 215$
(C) $p = 215$
(D) $p \geq 215$
(E) $p \leq 215$
- Q8.** The score s of a gamer in an online tournament is constrained by $s - 10 < 50$. What is the solution to this inequality?
- (A) $s < 60$
(B) $s > 60$
(C) $s = 60$
(D) $s \geq 60$
(E) $s \leq 60$

Open Ended Questions (FRQ)

- Q9.** A gaming company has the inequality $3x + 5 > 20$ representing the amount of data x (in GB) a user can download. Solve this inequality and express the solution set in interval notation. Provide your answer in the space below.
- Q10.** A coder is trying to optimize the speed of their game by solving the inequality $2t - 4 < 16$, where t represents the processing time (in milliseconds). Solve this inequality and graph the solution set on a number line. Provide your answer in the space below.

Chef's Tips

- Please ensure students have pencils, calculators, and erasers before starting the test.
- Encourage students to read each question carefully and manage their time wisely.
- Remind students to clearly label their answers and show their work where necessary.